

30 Affordances and Attention

Learning and Culture

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Abstract: *In this chapter, we draw on Gibson's (1979) description of affordances to consider cultural differences in motivation and learning. We develop the argument that affordances are at the heart of cultural differences. We address the way culture influences both what and how people learn from the affordances that are available to them in their physical and social environments. Brain processes of neural plasticity and psychological learning mechanisms of repetition and connection drawn from the Unified Learning Model (Shell et al., 2010) are used to explain how our brain and memory store knowledge of affordances, as well as the actions needed to take advantage of these affordances. We then discuss the way attention sits at the intersection of motivation and learning, as well as how motivated attention leads to individual and cultural differences in knowledge and use of affordances, both implicitly and volitionally. Finally, the emergence of cultural differences in attention, learning, knowing, and motivation are discussed, with an emphasis on the impact of culture on learning in school.*

Simon (1969) once noted that if you wanted to know what path an ant would take across the beach, you would know more from studying the beach than from studying the ant. This reflects Gibson's (1979) notion of affordances: "The affordances of the environment are what it offers the animal, what it provides or furnishes, either for good or ill" (p. 127). Back to Simon's ant, the terrain of the beach has specific hills and valleys. These characteristics of the beach terrain "afford" the ant the opportunity for some movements, but also, exclude doing other movements. Within the affordances of a specific environment, there also is the possibility of variation in the way different ants will move. If we were to observe a group of ants walking across the beach, it would immediately become apparent that although all are traversing the same beach, all are not traversing it in exactly the same way. There is a finite set of affordances available on the beach, yet individual ants are not necessarily experiencing all of them in exactly the same manner. There may be considerable variation in what each ant "knows" about the beach and in the exact movements used to traverse it. Thus, as noted in more recent ecological psychology theories (e.g., Chemero, 2003; Stoffregen, 2003), affordances involve

an interaction between the organism and the environment. The environment may constrain what is possible to do or to learn, but the organism's characteristics also constrain the specific affordances it can take advantage of. The beach may be the same beach, but the affordances on the beach are not only different for different ants, they are also different for you and me.

In this chapter, we draw on Gibson's (1979) description of affordances to anchor our examination of cultural differences in motivation and learning. We argue that Gibson's conception of affordances is at the heart of cultural differences. The environment affords possibilities that may be perceived and acted on, and people living in different physical environments perceive and experience different affordances. Just consider the environmental differences between the Amazon rain forest and the Sahara desert. The people in these environments have different affordances available. Just as physical environments provide varying affordances, so too social environments provide differing affordances. Consider the different experiences of majority and minority group residents of the same city. The affordances available in these social environments can be quite distinct, even if these individuals are sharing a similar physical environment. Such sociocultural differences are reflected in religions, stories, and transmitted knowledge with a culture. Moreover, each individual in the group has unique personal experience and knowledge derived in part from the affordances each has perceived and taken advantage of in the past. This unique personal knowledge subsequently can affect the affordances they are able to use in the future.

The notion that individuals may take advantage of only some of the affordances available in an environment because of the individual's unique personal characteristics is at the heart of learning, especially the way motivation influences learning. Our goal in this chapter is to address how cultural differences affect motivation and learning, especially how they affect the way persons learn from the affordances available to them. To begin, we will discuss the neurological processes involved in learning and the psychological learning theory, the unified learning model (ULM; Shell et al., 2010), derived from these neurological processes, which explains how our brain and memory store knowledge of affordances and the actions needed to take advantage of these affordances. We then will discuss how attention sits at the intersection of motivation and learning, and is the determining mechanism for individual differences in knowledge and use of affordances. Finally, we examine the emergence of cultural differences in learning, knowing, and motivation, with an emphasis on learning in school.

Learning

The Neurobiology of Learning

Neural plasticity is the basic underlying process of brain and neurological learning and development (Kandel et al., 2012). Neural plasticity refers to the

changes that occur in the neural connections of the brain at both the physical and functional levels. These changes occur both in the interconnections of neurons, through the creating and pruning of synapse connections between neurons, and in the efficiencies of existing connections, via changes in myelination and neurotransmitter properties at synapse junctions. Although the specifics of the biological mechanisms involved in neural plasticity are not fully known, new research is beginning to detail them (e.g., Stuchlik, 2014). However, the basics of neural plasticity mechanisms have been known for some time. Hebb (1949) was the first to articulate the basic principle of neural plasticity: associations between neurons that fire together are strengthened and associations between neurons that don't fire together are weakened. Subsequent neurological research has confirmed that Hebb's simple "use it or lose it" principle accurately describes how neural plasticity occurs in the brain (see Caporale & Dan, 2009; Kandel et al., 2012; Shors, 2014). Neurons that are active at the same time have their connections created or strengthened. Neurons that are not active at the same time are not connected or, if connected, have their connections weakened.

Neural plasticity accounts for the specific interconnections of neurons in the brain that change over time and are typically described as brain development. However, it is more accurate to describe neural plasticity as learning. Neural plasticity is not maturational or age graded. Neural plasticity is driven by stimulation. Neurons activate or fire in response to stimulation. Stimulation comes from sensory input and from internal activation via proprioceptive feedback, the sympathetic and parasympathetic regulatory systems, and emotions. Finally, stimulation also comes from retrieval of memories or motor neuron activation when these are activated by pattern matching to incoming stimuli or by the aforementioned internal activators.

We can think of the connections between neurons as recording the history of their repeated co-stimulation (see Corbetta, 2012). The strength of the connection reflects the number of times the neurons have been stimulated together (thereby having their connection strength increased) versus the number of times the neurons have been stimulated separately (thereby having their connection strength decreased). This historical record is biologically encoded in the brain via processes of long-term potentiation (LTP), synapse formation, synapse pruning, myelination, and changes in neurotransmitters at the synapse junction (Blumenfeld-Katzir et al., 2011; Driemeyer et al., 2008; Kandel et al., 2012; Miendlarzewska et al., 2016; Stuchlik, 2014).

The Unified Learning Model

Neural plasticity provides a mechanism for the brain and nervous system to perform two critical survival functions: (1) to allow accurate perception and understanding of the world around us and (2) to allow functional behavior in that world, presumably so that the organism can stay alive and reproduce

(Shell et al., 2010). These two functions are reflected in Gibson's (1979) definition of affordances. To survive, an animal must be able to perceive the affordances available and know what action to take to utilize them. Hence the need for an organism to have a way to store and remember affordances, and remember how to act to take advantage of known affordances. By biologically encoding a historical record of neuron co-stimulation, neural plasticity allows neuron connections to record what the organism knows about the world around it from past sensory input and what behaviors have been successful for the organism in meeting survival goals.

The unified learning model (Shell et al., 2010) translates these processes of Hebbian neural plasticity into two cognitive learning mechanisms: (1) *repetition*, the repeated exposure to sensory input, bodily sensations, retrieval of knowledge from long-term memory, or personal experience for declarative knowledge and repeated practice for procedural knowledge and (2) *connection*, the formation of larger units of meaning such as concepts or schema for declarative knowledge and development of stimulus-motor response patterns leading to automaticity for procedural knowledge. Shell et al. argue that repetition and connection are able to fulfill the aforementioned survival processes because of statistical learning. The strengthening and weakening of a neural connection each time the connection is repeated or not repeated can be represented as frequency counts. The statistical law of large numbers then says that as the number of repetitions increases, the frequency distribution will come to reflect the accurate probability distribution of whatever was being counted. In the case of neuron connections, these repetitions reflect the affordances that the organism has experienced. So as the organism experiences repeated affordances, their understanding of their world will become more accurate and their behavior will become more functional and successful. At the psychological level, repetition will ultimately produce declarative knowledge, which accurately reflects the probability distribution of whatever knowledge is being represented in a concept/schema and procedural knowledge, which accurately reflects the probability of successfully producing the desired behavior in a given context.

Numerous studies have shown that statistical learning occurs at the neurological level (e. g., Kirkham et al., 2002; Mitchel & Weiss, 2011; Schapiro et al., 2016; Turk-Browne et al., 2008). Studies have also shown that the learning of declarative knowledge and procedural knowledge follows statistical regularities (Kóbor et al., 2017). Language learning especially has been shown to follow statistical learning (Erickson & Thiessen, 2015). As noted by Shell in his contribution to Wang et al. (2013), statistical learning by neuronal connections allows each member of a cultural group, with their individually unique experience, to tend toward similar shared knowledge representations and similar recognizable behavior patterns. This is because, within the same physical or social environment, every person's unique experiences are bound by the physical and social affordances available in the shared environment. Hence, they are all moving toward the same statistical regularities present in

the affordances of that environment. This provides a basic explanation for cultural differences in knowledge and understanding. The shared environment differs across cultures, regions, and peoples. So, within each specific culture, the members will tend toward a shared understanding of their experienced environment but, because these experienced environments differ across cultures and peoples, members of different cultures will have different declarative and procedural knowledge.

Cognitive Processing Limitations on Learning

Statistical learning through the law of large numbers can explain how persons within a shared physical and cultural environment come to have similar knowledge. If neural plasticity was a passive process driven from the environment, something like direct perception of affordances (see Chemero, 2003; Stoffregen, 2003), then ultimately members of a shared environment would have little individual variation. Mathematically, the law of large numbers says that all should be moving toward the same statistical representations given similar amounts of repetition (e.g., Kolmogorov, 1950). Individual differences would be almost solely a function of variation in the number of repetitions. This clearly is not true. Although, with repetition, knowledge may be tending toward an accurate probability distribution of affordances, there can be considerable individual difference in knowledge among the members of the same cultural group despite shared affordances in their common social and physical environments (e.g., Na et al., 2010).

Individual differences occur because humans cannot directly perceive the totality of the affordances in their environment. Miller's (1956) well-known "Magical Number 7 ± 2 " was an early exploration of capacity limitations on information processing, immediate memory (short-term memory), and attention span. Subsequent studies of working memory (Baddeley, 1986; Cowan, 2010) and cognitive load (Sweller et al., 2011) have substantiated that limitations and constraints on what can be attended and processed exist at all levels of cognitive processing. Basically, the affordances available at any one time contain far more information than could possibly be attended, perceived, and processed.

These processing limitations have a profound effect on statistical learning. Shell et al. (2010) note the particularly critical role of the third learning mechanism proposed in the ULM: attention. Because of capacity limitations, attention can focus on only a sample of the possible information available. This means that on any given encounter with the environment, attention selects only a sample of the information available in affordances. So, even though all members of an environment may share the common available affordances, they are each sampling uniquely from the information available in those affordances. These individual differences in what they attend mean that each person's knowledge of affordances, and their neural representations of these affordances, are not

updating based on the totality of information in the affordances. The person's knowledge and neural representations are updating based only on the information that was in their individually attended samples. Sampling theory and the law of large numbers dictate that, mathematically, all combinations of unique samples would still tend toward the same probabilities in the affordances. If attended samples of affordances were random, which might occur if neural plasticity was a passive process of recording sensory input, then there would be little difference between updating based on all information available versus samples. All combinations of unique samples would ultimately produce identical encoding of the information available in the affordances. But, attended samples are not random – they are biased by the individual's unique knowledge, resulting from their past history of learning from experienced affordances.

Attention, Learning, and Motivation

Attention and Learning

To understand the way attention affects learning, we need to revisit neural plasticity. The actual biochemical processes of neural plasticity in the brain and nervous system are not under any sort of volitional control. One cannot tell a dendrite to grow or a nerve to myelinate. As we have discussed, these processes occur due to neuron stimulation resulting from repetition and connection. And one cannot directly tell a neuron to repeatedly stimulate or connect to another neuron. One can, however, volitionally direct cognitions and behaviors that will indirectly trigger neuron stimulation.

Within the psychological and educational literature, there is a long history of research into and identification of cognitive processes and behaviors that have been shown to influence learning and subsequent achievement in educational settings. These include, among others, self-regulated learning (e.g., Boekaerts & Cascallar, 2006; Schunk & Zimmerman, 2013), study/learning strategies (e.g., Weinstein et al., 2000), knowledge building (e.g., Bereiter & Scardamalia, 2014), metacognition (e.g., Brown et al., 1983), and engagement (e.g., Sinatra et al., 2015). These behaviors and cognitions all serve to trigger ULM learning mechanisms. General learning strategies (such as note-taking, re-reading, summarizing, and paraphrasing) all involve repetition. Sometimes these are literally repetition as in rehearsal strategies. The more sophisticated strategies, however, also primarily serve to either revisit or retrieve information multiple times or in multiple formats. Knowledge-building and other deep learning strategies (such as elaboration and organization, as well as summarizing) serve the ULM mechanism of connection. These strategies interconnect multiple pieces of learned information with strategies (like matrix notes) attempting to make connections overt (e.g., Jairam et al., 2012). From the perspective of the ULM, these strategic

learning methods are ways, at the cognitive and behavioral level, to indirectly trigger the neural processes involved in Hebbian neural plasticity by making students repeat and connect information.

More detailed examination of these strategic learning methods, however, indicates that they all involve attention. At both the neurological and cognitive level, repetition and connection are processes within the cognitive architecture and neural system. As discussed in the ULM (Shell et al., 2010), if information is attended, it is maintained in active cognition or working memory. In the brain, attention triggers processes such as LTP that maintain neuron activity (Kandel et al., 2012). Maintaining the information allows the information to be repeatedly attended. This retention is automatic once we attend to the information. We don't decide to retain the information or not; we only decide to attend. Similarly, attended information triggers automatic pattern-matching retrieval from long-term memory. So we will retrieve information from memory if it matches anything in current attention. This constitutes another form of repetition. Again, we do not decide to do pattern-match retrieval; we only attend.

Similarly, if two pieces of information are attended together, they will be automatically associated and stored as connected information. This is a function of the well-known learning mechanism of contiguity (e.g., Matzel et al., 1988; or more recently, Mayer & Fiorella, 2014), which reflects the Hebbian neural learning principle of "fire together; wire together." Subsequently, if either of the connected pieces of information is later attended, both will be retrieved from memory (or stimulated in the neurological system); thus their connection is repeated, even if only one was attended. Again, this is an automatic process. We do not decide to connect information, we only attend to information that is simultaneously in active cognition and working memory.

It is clear from the workings of these neuro-cognitive processes that although the ULM specifies three mechanisms of learning (attention, repetition, and connection), attention is the only one that is under volitional control. What we attend dictates whether repetition and connection occur at the neuro-cognitive level. Thus, we can reconceptualize the behaviors and cognitions subsumed under strategic learning, knowledge building, self-regulation, engagement, and studying as attention directing activities. Basically, they are ways, at the macro level of cognition and behavior, that we make ourselves focus our volitional attention. When I tell myself to "repeat" something, I am making myself attend. When I make a matrix outline, I am making myself attend to connections. When I relate something I am learning to prior knowledge, I am bringing the new information into contiguity with what I already know to make myself attend to both together.

Motivation and Attention

Because of the role attention plays in determining the information from available affordances that is sampled and learned, the cognitive and neurological

factors that influence attention play a significant role in learning. In the ULM, Shell et al. (2010) expanded on the premise first proposed by Brooks and Shell (2006): the role of motivation in learning is to direct attention in working memory (see also Anselme, 2007 for a similar view). There are multiple ways that attention and motivation intersect.

Motivation and attention in System 1. Motivation and attention first intersect in what Kahneman (2011) called “System 1.” Cognitive processes in System 1 include those that are part of the neurological and cognitive architecture. System 1 processes are not under volitional control. We have noted some of these before, such as the neurochemical processes of neural plasticity and basic biological functioning. Attention is directed automatically by System 1 processes, such as internal homeostatic drive mechanisms like thirst or hunger, and other proprioceptive and internal neural stimulation, such as pain (Horvitz, 2000). These automatic attention directors play critical roles in survival and underlie much classical and operant conditioning (Schultz, 2015). Anselme (2007) notes these as causal factors of the motivation that drives anticipatory attention. They direct attention toward those behaviors that can alleviate the drive by, say, providing food or water. These internal attention mechanisms are motivated through the primary System 1 motivator: the dopaminergic “reward system” (Hidi, 2016; Horvitz, 2000; Schultz, 2015). The reward system “computes” the value of possible environmental rewards, the prediction error of whether attained reward matches expected reward, contingency information for reward signals, and salience of stimuli (Bromberg-Martin et al., 2010; Hidi, 2016; Yin et al., 2008). Basically, the reward system provides a motivational boost for attending to those affordances that can most successfully reduce the biological drive.

Also, at System 1, attention can be directed automatically by aspects of the sensory environment, especially novelty and salience (see Knudsen, 2007). This is where the environmental aspects of affordances come into play. Physical environments have different salient attributes that need to be attended to for survival. These are the properties of the “beach” in Simon’s (1969) example. Different beaches will make certain physical properties of terrain, temperature, and so forth salient, and the regularities of the environment dictate what is novel or out of the ordinary. Because of the critical importance of attending to the environment for survival, the brain has evolved automatic mechanisms for directing attention to salient and novel sensory inputs (Knudsen, 2007). Motivationally, this “bottom-up” automatic attention also interacts with the dopaminergic “reward system” (see Bromberg-Martin et al., 2010; Horvitz, 2000; Miendlarzewska et al., 2016; Schultz, 2015; Yin et al., 2008), which can provide a motivational boost to attend to particular inputs, especially novel and salient inputs that meet the needs of biological drive.

These System 1 attention mechanisms are impacted by emotion, which is distinct from the dopaminergic reward system but interacts with it. Emotion’s role appears to be providing subtlety to rewards, especially degrees of pleasure

or pain (Vuilleumier, 2005, 2015) that are distinct from the information provided by the reward system. Although not as well understood as the reward system, emotion appears to affect attention through the amygdala and its projections to the prefrontal cortical areas (Vuilleumier, 2005, 2015).

Also in System 1, there is evidence that neurons compute confidence estimations in decision-making (Kiani & Shadlen, 2009) and that confidence is encoded in neuron firing rates as part of the attractor networks in decision-making, with multiple confidence networks allowing decision monitoring and change (Deco et al., 2013). These confidence networks interconnect with the reward system to compute confidence in receiving rewards (Deco et al., 2013). In a revision to the original ULM (Shell et al., 2010), Shell et al. (2015) proposed, and computationally modeled, self-efficacy as being integrated with knowledge itself in memory. They proposed that self-efficacy modulates connection strength as a function of certainty in the accuracy of knowledge or the likelihood of success at attaining outcomes. This self-efficacy modulation mirrors how neuron firing rate modulates the strength of neural connections. This self-efficacy modulation biases attention toward those actions with the best chance at being successful in attaining desired rewards.

Motivation and attention in System 2. These aforementioned System 1 attentional processes serve the two primary functions of the brain. Novelty and salience serve to direct attention to potentially important aspects of the environment that are likely to have significance for understanding the world and for survival. Homeostatic drive mechanisms serve to direct attention to biological survival needs and motivate action to alleviate these needs. The motivation from the dopaminergic reward system, emotion, and confidence estimation serve to motivate these immediate attention directors. However, we are not motivated only by the immediate situation.

We can only store what we attend to, primarily that which is salient and novel. The information that was attended was linked to its value and utility for alleviating drive mechanisms via the reward system (Berridge & Kringelbach, 2008; Bromberg-Martin et al., 2010) and, perhaps, emotion (Vuilleumier, 2005, 2015). The initial tagging of these environmental features by the reward system makes them more likely to be noticed and attended in the future. Behaviors that were successful at attaining reward (reinforcement) are strengthened and associated with the environmental stimuli signaling that they should be done. Thorndike (1913) called this the “law of effect.” The connections between stimuli and behaviors leading to reward are strengthened each time the behavior is successful in attaining the reward. Because of statistical learning, these connections will ultimately come to reflect the relative probability of success of the behavior in attaining the reward versus its failure.

These interactions of the reward system with learning anchor the first group of cognitive or knowledge-based motivators. Reward or reinforcement requires that we remember two things: first, the contingency of the reward on behavior; second, the magnitude or amount of the reward or its value. These

two aspects of the dopaminergic reward system are reflected in social learning/cognitive theories (Bandura, 1986; Rotter, 1966; Weisz & Stipek, 1982) and in control theories of behavior (Skinner, 1996) as outcome expectancies (contingency) and value. In social learning/cognitive theories, outcome expectancies and value are seen as the memory record of past experienced rewards/reinforcement. Looked at in another way, they are the memory record of past activation of the reward system. Expectancies and value are motivating because they contain information about the likelihood that a behavior will lead to a desired outcome. Rotter's (1966) locus of control construct was theoretically conceptualized as the generalization of specific expectancies to produce an overall belief that rewards/reinforcement were contingently related to one's behavior (internal locus of control) or not (external locus of control). Similar notions of contingency are present in future time perspective (FTP) theories in the constructs of instrumentality (which reflects the relationship of present behaviors to achieving future goals) and connectedness (which, like locus of control, reflects a general belief that the future is contingently connected to the present [Husman et al., 2015; Husman & Lens, 1999; Husman & Shell, 2008]). FTP theories also recognize a future valence aspect, similar to value, that reflects the valuing of future goals/outcomes relative to present goals/outcome (Husman & Lens, 1999; Husman & Shell, 2008). It is the memory record of contingency, and the magnitude of experienced reward reflected in outcome expectancies and their associated value, that allows for computation of prediction error by the reward system for discrepancies from the reward expected. Note that this tight coupling of cognitive motivators to the reward system informational components of contingency and reward value in social learning/cognitive and FTP theories are not present in what has been termed modern expectancy-value theory (Eccles & Wigfield, 2002). Modern expectancy-value theory has different definitions of expectancy and value that do not correspond directly to these reward system informational components.

The reward system and its cognitive analogs, outcome expectancy and value, provide information on contingency of behavior to rewarding outcomes and the value (magnitude) of the contingent reward. As Bandura (1997) notes, however, this does not mean that one is capable of doing the contingent behavior. Bandura (1986, 1997) argued that having a strong outcome expectancy for a high value reward is not motivating if one does not believe one could successfully do the contingent behavior. He called this belief self-efficacy, which is the confidence one has in being able to successfully produce a cognitive or overt behavior. Research examining self-efficacy and outcome expectancies together (Shell et al., 1989, 1995) has supported Bandura's contention that self-efficacy is more strongly predictive of achievement than outcome expectancy. As noted previously, there is some evidence that self-efficacy has an actual neurological underpinning (Deco et al., 2013).

Emotions also have a memorial component. Just like the information about contingency and value connected to knowledge by the reward system, emotions

connect nuances surrounding pleasure and pain to knowledge (Vuilleumier, 2005, 2015). Emotions are associated with stored information and impact the way memories are retrieved (Pekrun, 2006). Pekrun (2006) has theorized and found evidence to support the idea that emotion, expectancy, and value are interconnected at the cognitive level, mirroring interconnections between emotion and the reward system at the neurological level. He and his colleagues have also found that emotion influences attention in early sensory processing (Kuhbandner et al., 2011).

Interest has been described, similarly to affordances, as an interaction between the person and environment (Hidi & Renninger, 2006). Interest has also been shown to affect attention (Hidi & Renninger, 2006; Renninger & Hidi, 2016). In relation to affordances, situational interest derives from environmental novelty or salience (Renninger & Hidi, 2016); thus, it may serve to provide a motivational boost to sustaining attending to these aspects of the environment, perhaps by tapping the dopaminergic reward system or emotions (Renninger & Hidi, 2016). Interest itself, however, would seem to be distinct from the specific informational components in the neural reward system of value, contingency, and prediction error. At the cognitive level, individual interest is clearly distinct from the cognitive representations of reward as outcome expectancies and value. Individual interest may reflect the history of that which has been experienced as interesting (Hidi & Renninger, 2006). Like other top-down motivators, personal interest may then boost attention by recruiting activation of the reward system or emotions.

Finally, humans and other animals are goal directed or, as Tolman (1932) called it, purposive. Some goals are directed “bottom up” by System 1 level homeostatic drive mechanisms (e.g., goals to get food when hungry) or by sensory inputs (e.g., goals to explore novel stimuli). As Bandura (1986, 1997) noted, however, people have personal agency. We are active, volitional directors of our behavior and pursue our own personal goals. We are not just passive responders to System 1 motivators. Volitional goals exist in what Kahneman (2011) called System 2, or volitional, cognition. Most of these volitional goals are divorced from basic biological needs. In the ULM (Shell et al., 2010), goals motivate attention because we are motivated to do what is necessary to achieve the goal. Hence, we direct attention to that which will move us toward the goal. The other previously discussed cognitive motivators can be viewed as supporting goal directed attention. Value tells us how much reward is associated with the goal, expectancy tells us what behaviors are contingently related to the goal, and self-efficacy tells us the probability that we can successfully do the contingent behaviors. Emotion tells us how pleasant achieving the goal will be. Although the retrieval of expectancies, values, and self-efficacy from memory may be automatic by pattern matching to current active or working memory contents, these motivators do not necessarily produce automatic behavior. They are used for decision-making, to decide between alternative goals and alternative ways of achieving chosen goals. This allows volitional control of directing and sustaining attention in the face of competing motivators.

Social Learning and Attention

We have discussed the learning of affordances and motivation primarily from the perspective of direct experience. Humans, however, learn not only from their own personal experience, but also from other people. Social learning includes vicarious experience from observing others (Bandura, 1986, 2016) and from verbal interaction (Bandura, 1986; Vygotsky, 1980). Bandura (1986) has demonstrated that vicarious learning leads to the same progression of skill as actual practice, indicating that vicarious experience triggers the same neural statistical learning as direct behavior. When learning vicariously, people do not just learn procedural knowledge or behavior, they also learn expectancies, value, and self-efficacy from observation. Again, the effects appear to be identical to having actually received the reward or been successful themselves.

Berridge and Kringelbach (2008) have noted that much of what we find “intrinsically motivating” has no real connection to basic System 1 survival motivators in the reward system. Although a learned motivator such as money could arguably be seen as a means to procuring something like food that would alleviate a homeostatic drive, many things that we find enjoyable and motivating have no connection to any sort of survival function. For example, we have learned to find things such as art rewarding for its own sake. Similarly, the associated emotional feelings of pleasure have no basis in any physical property of the stimulus or activity; we have learned to feel the emotional reactions. So where did we learn these motivators? We learned them from other people.

That humans are social animals has long been noted. Various genetic bases for social affiliation have been proposed in evolutionary psychology (e.g., Seyfartha & Cheney, 2013). Any genetic underpinnings might be expected to have produced neurological motivating mechanisms for social functions and social learning, but the research into these is in its infancy (e.g., Feldman et al., 2016; Harmon-Jones & Inzlicht, 2016; Numan, 2014; Striepen et al., 2014). So, the neurological basis for motivation and attention in social functioning has yet to be fully identified. Social interactions do appear to trigger reactions in the dopaminergic reward and emotion systems (Striepen et al., 2014), suggesting that social interactions may receive motivational boosts from those that trigger attention to socially relevant people and behaviors. But, social interactions also appear to direct attention independently of the reward system through other neural mechanisms such as the oxytocin pathway (Feldman et al., 2016; Striepen et al., 2014). It is clear that social cognitions direct attention.

In relation to affordances, for humans, the social/cultural environment often provides more affordances than the physical environment. It is true that many people's primary concerns are meeting basic survival mechanisms. For these people, social affordances tend to be heavily focused on environmental affordances that fulfill basic needs. But, in technologically advanced societies, social affordances can be quite different from the affordances in the environment and can be focused on socially constructed objects, behaviors, rewards, and motivations. For these people, the balance of the interaction between the person and the environment in affordances tilts toward the person.

Regardless of the focus of these social affordances, the majority of knowledge that humans learn is what Vygotsky (1980) called *socially constructed*. This knowledge is not a property of the physical environment; rather, it is a property of the social environment. Parents and other adults direct children's attention, and emphasize and repeat what is important. They model behaviors, tell stories, and introduce the child to cultural and social traditions and institutions (like religion). They explain the meaning of stimuli and actions in the world. They explain the meaning of feelings. These types of social knowledge transmissions are carried on by teachers in formal schooling. Our understanding of affordances in the physical environment is filtered through these socially constructed meanings. Others in our social group have guided our attention in ways that dictate the content of the affordance samples that we are attending.

The effects of socially learned, socially directed, and what might be called socially filtered affordances on attention are profound. Although we have been discussing motivation and attention primarily from the “bottom-up” System 1 perspective, the effects of “top-down” processes are even greater (Knudsen, 2007; Sarter et al., 2001). As discussed by Shell et al. (2010) in the ULM, incoming sensory information (and internal stimuli from homeostatic drive, proprioceptive, or pain) triggers retrieval of stored knowledge from long-term memory. These incoming stimuli trigger attention, but the activated long-term memory knowledge can reinforce or override the incoming stimuli and, thus, either sustain or redirect attention. This long-term memory knowledge was initially tagged with reward values or emotions, and is connected in memory with cognitive motivators. As the knowledge is activated, these associated motivators are also activated. Neurologically, retrieved information can stimulate the dopaminergic reward system and emotions via top-down neural projections into brain areas associated with the reward system and emotion (Bourgeois et al., 2016; Sarter et al., 2001). As discussed by Anderson (2016), retrieved knowledge that was previously rewarded can compete for attention even when irrelevant to the current task. These top-down influences have been called the prior knowledge effect (Shell et al., 2010). Existing knowledge biases attention to stimuli that correspond and reinforce current knowledge. This creates a self-fulfilling loop in which people attend more to information consistent with current knowledge, thus increasing the strength of that knowledge. Piaget (1971) noted this process as assimilation; we assimilate new information by incorporating it into existing knowledge. Thus, differences in culture-based, socially constructed knowledge cause perceptions of particular affordances to differ.

Culture, Affordances, and Learning

Park and Huang (2010) note that there is growing evidence in support of cultural influences on neural functioning. These influences are especially evident with regard to activation of the ventral visual cortex, an area associated with perceptual processing, including attention. Consistent with the ULM (Shell et al., 2010), Park and Huang note that the method of culture-based

knowledge transfer – in the form of repeated experiences, thoughts, and perspectives – influences the way in which neural pathways get wired and are activated. The prior knowledge constructed from these experiences exerts top-down influences on attention. Sometimes referred to as “ways of knowing” (Warner, 2006), these well-established patterns influence the way members of a culture interpret and assimilate new information and, ultimately, the long-term memory knowledge that is created. These ways of knowing constitute social/cultural affordances and provide socially constructed meanings for interacting with physical affordances in the environment.

For example, many indigenous cultures understand the physical world as living parts of a larger whole. In Western thought, animals and plants are alive in different ways, and rocks are not alive. Many indigenous cultures understand the rocks to be alive, the waters to be alive, and the clouds to be alive: those are fundamentals taught from childhood. Furthermore, all things are related. Nothing exists in isolation; things done to one aspect of creation are done to every aspect. The connection is distinct. Much like the Western tradition of discrete categorization, it is the goal of science to organize, categorize, and isolate variables as a means of better understanding all the parts. People from Western cultures direct attention to focal points – key aspects of phenomena. However, people from indigenous cultures may attend to contextual factors. These basic perceptual functions influence the way learning takes place. It seems clear that the affordances available in the environment can be very different for indigenous and non-indigenous people.

Deloria and Wildcat (2001) speak to this issue. The identity and perspectives of Native American students are inextricably linked to places of ancestral culture and experience. They explain that “there is no such thing as isolation from the rest of creation, and the fact of this relatedness provides a basic context within which education in the growth of personality and the acquisition of technical skills can occur” (p. 45). They emphasize that “indigenous people represent a culture emergent from a place, and they actively draw on the power of that place” and “behavior, beliefs, values, symbols, and material products” are emergent from these diverse places (p. 31). These are powerful social/cultural affordances.

Clearly, differences in cultural experience, and the corresponding allocation of attention and practiced behavior, shape learning and knowledge development. Mooney et al. (2016) confirm that “our cultural experiences, beliefs, and values play a large part in whom we are, how we see ourselves, and what we think is important” (p. 13). And, for Aboriginal students, “identity is influenced by their environment and reinforced by significant others, including the broader Aboriginal community” (p. 13). Emotion, as displayed in facial expressions, is not universal. Culture influences the way emotion is expressed on the face and how it is interpreted by others (Jack et al., 2009). The communication of emotion, therefore, may be misinterpreted among persons from different cultural backgrounds. Indigenous communities maintain a shared

identification with a specific land or place. Indigenous students may “hold beliefs, have perceptions and experiences, and design goals that are different from mainstream, majority educators” (McInerney & Flowerday, 2016, p. 1). Relationships, community, and place will likely be themes. Cajete (2015), a leading Native American scholar, reminds us that “relationship, respect, and responsibility are primary indigenous values, because they lay the foundations for positive communal processes” (p. 64).

Recall that affordances involve interaction between the person and the environment. These cultural differences lead to dramatically different affordance interaction. When traveling across the northern plains of South Dakota, a tourist might be noticing the flat, grassy landscape, the seemingly endless blue sky, and the occasional rocky butte. They may be looking at road signs, noticing locations of nearby gas stations, or simply wondering when the next town might appear. An indigenous person traveling across the same grassy plain could be searching the sky for eagles, listening to the meadowlarks, giving thanks for the clouds moving in from the west, greeting buttes as relatives, and offering a prayer to a sacred location as it is approached. The connection to place is central for indigenous people, and a seemingly empty landscape can be alive with memories, stories, and spiritual relationships. The physical affordances of the environment are the same, but attention may be directed in different ways and, therefore, the experienced affordance is very different.

We can now use the lens of the ULM (Shell et al., 2010) to specify the mechanism for cultural influences on learning. To recap: the knowledge that we have stored in memory is the result of the learning of information available in physical and social affordances in our environment. We have learned this knowledge through repetition and connection of information that we have attended. Because of statistical learning, our knowledge tends toward accurate representation of the statistical regularities present in the affordances. But the statistical regularities in our knowledge reflect only the information in the samples of the information from the affordances that we have attended. What we have attended is a function of motivation. This motivation derives from internal biological and neurological processes and from social learning.

How different cultures define rewards affects which things people learn to pay attention to for rewards. That which is defined as beautiful or valuable is subjective and differing across cultures. But the effect on things people might think of as intrinsic properties of physical stimuli is perhaps even more profound. Take something like taste. At a basic chemical level, taste sensory organs can detect certain properties (e.g., sweet, bitter, salty). But the combinations of these that we find pleasing is a function of culture and the food we have learned to eat. People in different cultures find different foods pleasant and desirable, because each culture identifies different affordances in the food to be rewarding.

The cultural norms transmitted by social interaction produce a cultural sampling bias in any encounter with affordances in the physical or social

environment. These sampling biases affect the knowledge from available affordances that is repeated and connected, which leads to different cumulative probabilities in the person's knowledge of the world and expected successful behavior for attaining reward. Once learned, this culturally biased knowledge affects all future attention and biases all new attended samples, directing attention to those aspects of affordances that reinforce the existing knowledge.

Motivation, Culture, and Learning in School

When we consider motivation and learning in educational settings, it is clear that we cannot directly employ findings specific to what is known about the mechanisms of neural plasticity to practice. Recall that the goal of neural learning is to allow accurate understanding, and this is accomplished through statistical learning and storage of the history of repeated stimulations. However, for the brain to accomplish this, people need to interact with the world and extract information from available affordances. We cannot directly stimulate a neuron; we must obtain our information from the world.

As previously discussed, our perceptions of available affordances are filtered through social learning. Although much of the social knowledge and social interaction affecting perceptions of affordances occurs informally in homes and social groups, the main avenue for instruction about perceiving and responding to affordances is school. Formal schooling is especially important for societies in which the balance of knowledge has shifted heavily toward mathematics, science, and engineering (the so-called STEM fields), philosophy, and literature. As noted by Vygotsky (1980), the knowledge of these fields is not a property of the affordances of the physical environment; the knowledge is the property of the affordances of the social environment.

The affordances of the physical environment are not passively perceived directly, and neither are the socially determined affordances. Students in schools do not simply store the knowledge presented by teachers or react to the instructional materials provided. Students are active directors of their own attention, even if they are not always aware that this is the case. We draw on one of our recent studies to provide an illustration. We found that situational interest influenced reading time, which subsequently influenced learning (Flowerday & Shell, 2015). This could be interpreted, as situational interest often is, as an indication that some affordance in the reading material captured interest and, thus, directed attention and subsequent effort (indicating repetition and connection) to relevant information in the reading. But we know that attention to affordances is not solely a function of properties of the environment. Flowerday and Shell also found that topic interest (an indicator of knowledge-based or personal interest) influenced situational interest. This would reflect the System 2, top-down biasing of attention; the person's knowledge influenced whether they saw information in the text as interesting. In this example, we can see the interaction between the person and the environment

in affordances (Chemero, 2003; Stoffregen, 2003). The ability of the environment to capture interest is partly a function of what the person is predisposed to seeing as interesting, based on prior responses to the affordances of their environment. Given this, it is clear that cultural differences in personal knowledge might exert a considerable impact on affordances in the school environment that trigger situational interest and attention.

In the same way that the affordances in school are specialized, socially constructed knowledge, the motivators that direct student attention in school are generally not directly connected to any sort of basic System 1 survival motivators in the dopaminergic reward system, such as rewards leading to alleviating hunger. Affordances in school typically are not going to have any intrinsic motivation in relation to survival or biological pleasure. They are going to be rewarding or motivating because, as Berridge and Kringelbach (2008) discuss, they have been given value through social interaction and social learning. This means that the rewarding properties of school affordances must be transmitted by teachers and other adults, and learned by students. Prior cultural knowledge that students have acquired from their communities and parents, about important rewards and motivation, may be at odds with what the teachers and the school view as rewarding.

The classroom is its own environment and, as such, provides certain affordances. Within this classroom environment, however, students bring their own prior knowledge based on their own experiences. Through System 1 biasing, these prior knowledge effects may have more influence on attention than System 2 motivators that may be present in school. Mooney et al. (2016) found significant differences between Aboriginal and non-Aboriginal students on factors of cultural diversity, cultural identity, and Aboriginal perspective. Aboriginal students were less likely to demonstrate behaviors associated with engagement. Deloria and Wildcat (2001) note that in traditional tribal education, accomplishments are not seen as individual that are seen as the accomplishments of the family. Flowerday (2016) notes that when teaching motivation theories to students, it is critical to ask them to “relate motivation to their lived experiences so they can make connections to the theoretical constructs” (p. 119). For indigenous students, this often means relating ideas to experiences in their family or community. Mooney et al. (2016) found that Aboriginal students were less likely to demonstrate behaviors associated with engagement. Rubie-Davis and Peterson (2016) found that Maori students had higher levels of performance goal orientation than their European peers, perhaps because achieving at high levels brings honor to their families. Also, Maori students felt the need to outperform peers as a means of demonstrating their competence in an environment which underestimates their abilities.

Flowerday et al. (2018) found that, in typical elementary and middle-school classrooms, ethnic minority students need to have higher levels of System 2 motivated self-directed learning to achieve the same level of math performance as white students. This higher System 2 motivated self-direction may

be needed to overcome mismatches between their own prior cultural knowledge and the knowledge assumed by the classroom environment. Since the classroom environment is more compatible with white majority student prior knowledge, those students can be successful with less System 2 motivated self-directed learning than ethnic minority students, because they have more compatible System 1 automatic attention match between the classroom and their personal prior knowledge.

Concluding Thoughts

Although statistical learning is well established in many neurological (e.g., Kirkham et al., 2002; Mitchel & Weiss, 2011; Schapiro et al., 2016; Turk-Browne et al., 2008) and behavioral/cognitive (Erickson & Thiessen, 2015; Kóbor et al., 2017) studies, the argument by Shell et al. (2010) in the ULM that statistical learning is sufficient to explain all learning is untestable at present. The Hebb (1949) neural plasticity mechanisms of repetition and connection are well established as the neural processes of learning. But we cannot yet watch the specific biological processes at the neural level in real-time to see change after each repetition or to see these changes unfold across the life span. Also, neuroscience has only begun to examine how cultural experience affects the way information is processed by the brain.

Really getting at the heart of motivation and learning in school, as well as their neurological foundations, will require the examination of these processes in real classrooms with real content, which is beyond present technology. While we wait for new research technologies, we see a need to dig deeper into the processes of motivation and strategic self-regulation at the cognitive and behavioral levels. Our work has been looking at understanding how the combined and joint influences of motivation drive different patterns of strategic self-regulation (Nelson et al., 2015; Peteranetz et al., 2017; Shell & Husman, 2008; Shell & Soh, 2013). These studies employ profiling or person-centered methods. We have especially been using these to examine cultural and ethnic differences in profiles (Flowerday et al., 2018). We have also been trying to examine the dynamics of change in aspects of motivation across classes or school years to better understand how specific motivational constructs evolve, and how individual constructs might be associated with other aspects of motivated self-regulation profiles and differences in classroom environments (Flanigan et al., 2017; Peteranetz et al., 2018). In respect to cultural differences, we have been examining motivational beliefs among Native American students in the Southwest, and how these beliefs change from fifth to sixth grade.

Perhaps the biggest challenge is to somehow connect these macro-level efforts (that use primarily survey questionnaires) to more micro-level, fine-grained, real-time assessments that could potentially link our self-report and

behavioral assessments more closely to underlying cognitive and neurological mechanisms. Our initial forays into this (Eck et al., 2016) have found that student motivated self-regulation profiles (Nelson et al., 2015; Shell & Soh, 2013) could be linked to real-time student trace data during a Wiki assignment. The ULM learning processes of repetition (indicated by multiple viewing, editing, or commenting on the same page) and connection (indicated by viewing, editing, or commenting on multiple unique pages) differed for students adopting different motivated self-regulation profiles during the course. Furthermore, such efforts will be needed to move our understanding of the role of culture in motivation, learning, and attention forward.

References

- Anderson, B. A. (2016). The attention habit: How reward learning shapes attentional selection. *Annals of the New York Academy of Science*, 1369, 24–39. doi: 10.1111/nyas.12957.
- Anselme, P. (2007). Some conceptual problems with the classical theory of behavior. *Behavioural Processes*, 75, 259–75.
- Baddeley, A. (1986). *Working memory*. Oxford: Clarendon Press.
- Bandura, A. (1986). *Social foundations of thought and action: A social cognitive theory*. Englewood Cliffs, NJ: Prentice-Hall.
- Bandura, A. (1997). *Self-efficacy: The exercise of control*. New York, NY: W. H. Freeman/Times Books/ Henry Holt & Co.
- Bandura, A. (2016). The power of observational learning through social modeling. In R. J. Sternberg, S. T. Fisk, & D. J. Foss (Eds.), *Scientists making a difference: One hundred eminent behavioral and brain scientists talk about their most important contributions* (pp. 235–9). New York, NY: Cambridge University Press.
- Bereiter, C. & Scardamalia, M. (2014). Knowledge building and knowledge creation: One concept, two hills to climb. In S. C. Tan, H. Y. So, & J. Yeo (Eds.), *Knowledge creation in education* (Education Innovation Series). Singapore: Springer. doi: 10.1007/978-981-287-047-6_3.
- Berridge, K. C. & Kringelbach, M. L. (2008). Affective neuroscience of pleasure: Reward in humans and animals. *Psychopharmacology (Berl)*, 199(3): 457–80. doi: 10.1007/s00213-008-1099-6.
- Blumenfeld-Katzir, T., Pasternak, O., Dagan, M., & Assaf, Y. (2011). Diffusion MRI of structural brain plasticity induced by a learning and memory task. *PLoS ONE* 6(6): e20678. doi: 10.1371/journal.pone.0020678.
- Boekaerts, M. & Cascallar, E. (2006). How far have we moved toward the integration of theory and practice in self-regulation? *Educational Psychology Review*, 18(3), 199–210.
- Bourgeois, A., Chelazzi, L., & Vuilleumier, P. (2016). How motivation and reward learning modulate selective attention. *Progress in Brain Research*, 229, 325–42. doi: 10.1016/bs.pbr.2016.06.004.
- Bromberg-Martin, E. S., Matsumoto, M., & Hikosaka, O. (2010). Dopamine in motivational control: Rewarding, aversive, and alerting. *Neuron*, 68(5), 815–34. doi: 10.1016/j.neuron.2010.11.022.

- Brooks, D. W. & Shell, D. F. (2006). Working memory, motivation, and teacher-initiated learning. *Journal of Science Education and Technology*, 15(1), 17–30.
- Brown, A. L., Bransford, J. D., Ferrara, R. A., & Campione, J. C. (1983). Learning, remembering, and understanding. In J. H. Flavell & E. M. Markman (Eds.), *Handbook of child psychology* (4th ed., Vol. 3). New York, NY: Wiley.
- Cajete, G. A. (2015). *Indigenous community: Rekindling the teaching of the seventh fire*. St. Paul, MN: Living Justice Press.
- Caporale, N. & Dan, Y. (2009). Spike timing-dependent plasticity: A Hebbian learning rule. *Annual Review of Neuroscience*, 31, 25–46.
- Chemero, A. (2003). An outline of a theory of affordances. *Ecological Psychology*, 15(2), 181–95.
- Corbetta, M. (2012). Functional connectivity and neurological recovery. *Developmental Psychobiology*, 54, 239–53.
- Cowan, N. (2010). The magical mystery four: How is working memory capacity limited, and Why? *Current Directions in Psychological Science*, 19(1): 51–7. doi: 10.1177/0963721409359277.
- Deco, G., Rolls, E. T., Albantakis, L., & Romo, R. (2013). Brain mechanisms for perceptual and reward-related decision making. *Progress in Neurobiology*, 103, 194–213.
- Deloria, Jr., V. & Wildcat, D. (2001). *Power and place: Indian education in America*. Golden, CO: Fulcrum Resources.
- Driemeyer, J., Boyke, J., Gaser, C., Buchel, C., & May, A. (2008) Changes in gray matter induced by learning – Revisited. *PLoS ONE* 3(7): e2669. doi: 10.1371/journal.pone.0002669.
- Eccles, J. S. & Wigfield, A. (2002). Motivational beliefs, values, and goals. *Annual Review of Psychology*, 53(1), 109–32.
- Eck, A., Soh, L.-K., & Shell, D. F. (2016). Investigating differences in wiki-based collaborative activities between student engagement profiles in CS1. *Proceedings of the 47th ACM Technical Symposium on Computer Science Education (SIGCSE'2016)* (pp. 36–41). New York, NY: ACM. doi: 10.1145/2839509.2844615.
- Erickson, L. C. & Thiessen, E. D. (2015). Statistical learning of language: Theory, validity, and predictions of a statistical learning account of language acquisition. *Developmental Review*, 37, 66–108.
- Feldman, R., Monakhov, M., Pratt, M., & Ebstein, R. P. (2016). Oxytocin pathway genes: Evolutionary ancient system impacting on human affiliation, sociality, and psychopathology. *Biological Psychiatry*, 79(3), 174–84. doi: 10.1016/j.biopsych.2015.08.008.
- Flanigan, A. E., Peteranetz, M. S., Shell, D. F., & Soh, L.-K. (2017). Implicit intelligence beliefs of computer science students: exploring change across the semester. *Contemporary Educational Psychology*, 48, 179–96. doi: 10.1016/j.cedpsych.2016.10.003.
- Flowerday, T. (2016). Using motivation to teach motivation. In M. C. Smith & N. DeFrates-Densch (Eds.), *Challenges and innovations in educational psychology teaching and learning* (pp. 109–22). Charlotte, NC: Information Age Publishing.
- Flowerday, T. & Shell, D. F. (2015). Disentangling the effects of interest and choice on learning, engagement, and attitude. *Learning and Individual Differences*, 40, 134–40. doi: 10.1016/j.lindif.2015.05.003.

- Flowerday, T., Shell, D. F., & Moreno, R., (2018). Using profiles of motivated strategic self-regulation to understand mathematics achievement of ethnically diverse elementary school students. Manuscript submitted for publication.
- Gibson, J. J. (1979). *The ecological approach to visual perception*. Boston, MA: Houghton Mifflin.
- Harmon-Jones, E. & Inzlicht, M. (Eds.) (2016). *Social neuroscience: Biological approaches to social psychology*. New York, NY: Routledge, Psychology Press.
- Hebb, D. O. (1949). *The organization of behavior: A neuropsychological theory*. New York, NY: Wiley & Sons.
- Hidi, S. (2016). Revisiting the role of rewards in motivation and learning: Implications of neuroscientific research. *Educational Psychology Review*, 28, 61–93. doi: 10.1007/s10648-015-9307-5.
- Hidi, S. & Renninger, K. A. (2006). The four-phase model of interest development. *Educational Psychologist*, 41, 111–27.
- Horvitz, J. C. (2000). Mesolimbocortical and nigrostriatal dopamine responses to salient non-reward events. *Neuroscience*, 96(4), 651–6.
- Husman, J., Brem, S. K., Banegas, S., Duchrow, D. W., & Haque, S. (2015). Learning and future time perspective: The promise of the future-rewarding in the present. In M. Stolarski, N. Fieulaine, & W. van Beek (Eds.), *Time perspective theory: Review, research and application: Essays in Honor of Philip G. Zimbardo* (pp. 131–41). Springer International Publishing. doi: 10.1007/978-3-319-07368-2_8.
- Husman, J. & Lens, W. (1999). The role of the future in student motivation. *Educational Psychologist*, 34(2), 113–25.
- Husman, J. & Shell, D. F. (2008). Beliefs and perceptions about the future: A measurement of future time perspective. *Learning and Individual Differences*, 18, 166–75.
- Jack, R. E., Blais, C., Scheepers, C., Schyns, P. G., & Caldara, R. (2009). Cultural confusions show that facial expressions are not universal. *Current Biology*, 19(18), 1543–8. doi: 10.1016/j.cub.2009.07.051.
- Jairam, D., Kiewra, K. A., Kauffman, D. F., & Zhao, R. (2012). How to study a matrix. *Contemporary Educational Psychology*, 37, 128–35.
- Kahneman, D. (2011). *Thinking fast and slow*. New York, NY: Farrar, Straus, & Giroux.
- Kandel, E. R., Schwartz, J. H., Jessell, T. M., Siegelbaum, S. A., & Hudspeth, A. J. (2012). *Principles of neural science* (5th ed.). New York, NY: McGraw-Hill.
- Kiani, R. & Shadlen, M. N. (2009). Representation of confidence associated with a decision by neurons in the parietal cortex. *Science*, 324, 759–64.
- Kirkham, N. Z., Slemmer, J. A., & Johnson, S. P. (2002). Visual statistical learning in infancy: Evidence for domain-general learning mechanism. *Cognition*, 83, B35–B42. doi: 10.1016/S0010-0277(02)00004-5.
- Kóbor, A., Janacsek, K., Takács, A., & Nemeth, D. (2017). Statistical learning leads to persistent memory: Evidence for one-year consolidation. *Scientific Reports*, 7, Article number: 760. doi: 10.1038/s41598-017-00807-3.
- Kolmogorov, A. N. (1950). *Foundations of the theory of probability*. New York, NY: Chelsea Publishing.
- Kuhbandner, C., Lichtenfeld, S., & Pekrun, R. (2011). Always look on the broad side of life: Happiness increases the breadth of sensory memory. *Emotion*, 11(4), 958–64. doi: 10.1037/a0024075.

- Knudsen, E. I. (2007). Fundamental components of attention. *Annual Review of Neuroscience*, 30, 57–78. doi: 10.1146/annurev.neuro.30.051606.094256.
- Matzel, L. D., Hel, F. P., & Miller, R. R. (1988). Information and expression of simultaneous and backward associations: Implications for contiguity theory. *Learning and Motivation*, 19, 317–44.
- Mayer, R. E. & Fiorella, L. (2014). Principles for reducing extraneous processing in multimedia learning: Coherence, signaling, redundancy, special contiguity, and temporal contiguity principles. In R. E. Mayer (Ed.), *The Cambridge handbook of multimedia learning* (2nd ed., pp. 279–315). New York, NY: Cambridge University Press.
- McInerney, D. M. & Flowerday, T. (2016). Indigenous issues in education and research: Looking forward. *Contemporary Educational Psychology*, 47, 1–3.
- Miendlarzewska, E. A., Bavelier, D., & Schwartz, S. (2016). Influence of reward motivation on human declarative memory. *Neuroscience and Biobehavioral Reviews*, 61, 156–76.
- Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, 63, 81–97.
- Mitchel, A. D. & Weiss, D. J. (2011). Learning across senses: Cross-modal effects in multisensory statistical learning. *Journal of Experimental Psychology: Learning, Memory, & Cognition*, 37, 1081–91.
- Mooney, J., Seaton, M., Kaur, G., Marsh, H. W., & Yeung, A. S. (2016). Cultural perspectives on Indigenous and non-indigenous Australian students' school motivation and engagement. *Contemporary Educational Psychology*, 47, 11–23.
- Na, J., Grossmann, I., Varnum, M. E. W., Kitayama, S., Gonzalez, R., & Nisbett, R. E. (2010). Cultural differences are not always reducible to individual differences. *Proceedings of the National Academy of Sciences of the United States of America*, 107(14), 6192–7. doi: 10.1073/pnas.1001911107.
- Nelson, K. G., Shell, D. F., Husman, J., Fishman, E. J., & Soh, L. K. (2015). Motivational and self-regulated learning profiles of students taking a foundational engineering course. *Journal of Engineering Education*, 104(1), 74–100. doi: 10.1002/jee.20066.
- Numan, M. (2014). *Neurobiology of social behavior: Toward an understanding of the prosocial and antisocial brain*. Oxford: Academic Press.
- Park, D. C. & Huang, C. M. (2010). Culture wires the brain: A cognitive neuroscience perspective. *Perspectives on Psychological Science*, 5(4), 391–400.
- Pekrun, R. (2006). The control-value theory of achievement emotions: Assumptions, corollaries, and implications for educational research and practice. *Educational Psychology Review* 18(4), 315–41.
- Peteranetz, M. S., Flanigan, A. E., Shell, D. F., & Soh, L.-K. (2017). Computational creativity exercises: An avenue for promoting learning in computer science. *IEEE Transactions on Education*, 60(4), 305–13. doi: 10.1109/TE.2017.2705152.
- Peteranetz, M. S., Flanigan, A. E., Shell, D. F., & Soh, L.-K. (2018). Career aspirations, perceived instrumentality, and achievement in undergraduate computer science courses. *Contemporary Educational Psychology*, 53, 27–44. doi: 10.1016/j.cedpsych.2018.01.006.
- Piaget, J. (1971). *Genetic epistemology*. New York, NY: W.W. Norton.
- Renninger, K. A. & Hidi, S. (2016). *The power of interest for motivation and engagement*. New York, NY: Routledge.

- Rotter J. B. (1966). Generalized expectancies for internal versus external control of reinforcement. *Psychology Monographs*, 80, 1–28.
- Rubie-Davis, C. M. & Peterson, E. R. (2016). Relations between teachers' achievement over- and underestimation and students' beliefs for Maori and Pakeha students. *Contemporary Educational Psychology*, 47, 72–83.
- Sarter, M., Givens, B., & Bruno, J. P. (2001). The cognitive neuroscience of sustained attention: Where top-down meets bottom-up. *Brain Research Reviews*, 35, 146–60.
- Schapiro, A. C., Turk-Browne, N. B., Norman, K. A., & Botvinick, M. M. (2016). Statistical learning of temporal community structure in the hippocampus. *Hippocampus*, 26(1), 3–8. doi: 10.1002/hipo.22523.
- Schultz, W. (2015). Neuronal reward and decision signals: From theories to data. *Physiological Review*, 95, 853–951. doi: 10.1152/physrev.00023.
- Schunk, D. H. & Zimmerman, B. J. (2013). Self-regulation and learning. In W. M. Reynolds, G. E. Miller, & I. B. Weiner (Eds.), *Handbook of psychology* (Vol. 7, pp. 45–68). Hoboken, NJ: John Wiley & Sons.
- Seyfartha, R. M. & Cheney, D. L. (2013). Affiliation, empathy, and the origins of theory of mind. *Proceedings of the National Academy of Sciences of the United States of America*, 110 (Supplement 2), 10349–56. doi: 10.1073/pnas.1301223110.
- Shell, D. F., Brooks, D. W., Trainin, G., Wilson, K., Kauffman, D. F., & Herr, L. (2010). *The unified learning model: How motivational, cognitive, and neurobiological sciences inform best teaching practices*. Dordrecht, Netherlands: Springer.
- Shell, D. F., Colvin, C., & Bruning, R. H. (1995). Self-efficacy, attribution, and outcome expectancy mechanisms in reading and writing achievement: Grade level and achievement level differences. *Journal of Educational Psychology*, 87, 386–98. doi: 10.1037/0022-0663.87.3.386.
- Shell, D. F. & Husman, J. (2008). Control, motivation, affect, and strategic self-regulation in the college classroom: A multidimensional phenomenon. *Journal of Educational Psychology*, 100(2) 443–59.
- Shell, D. F., Murphy C. C., & Bruning, R. H. (1989). Self efficacy and outcome expectancy mechanisms in reading and writing achievement. *Journal of Educational Psychology*, 81, 91, 100. doi: 10.1037/0022-0663.81.1.91.
- Shell, D. F. & Soh, L.-K. (2013). Profiles of motivated self-regulation in college computer science courses: Differences in major versus required non-major courses. *Journal of Science Education and Technology*, 22(6), 899–913.
- Shell, D. F., Soh, L.-K., & Chiriacescu, V. (2015). Modeling self-efficacy as a dynamic cognitive process with the Computational-Unified Learning Model (C-ULM): Implications for cognitive informatics and cognitive computing. *International Journal of Cognitive Informatics and Natural Intelligence*, 9(3), 1–24. doi: 10.4018/IJCINI.2015070101.
- Shors, T. J. (2014). The adult brain makes new neurons, and effortful learning keeps them alive. *Current Directions in Psychological Science*, 23(5) 311–18.
- Simon, H. A. (1969). *The sciences of the artificial*. Cambridge, MA: MIT Press.
- Sinatra, G. M., Heddy, B. C., & Lombardi, D. (2015). The challenges of defining and measuring student engagement in science. *Educational Psychologist*, 50, 1–13, doi: 10.1080/00461520.2014.1002924.
- Skinner, E. A. (1996). A guide to constructs of control. *Journal of Personality and Social Psychology*, 71, 549–70.

- Stoffregen, T. A. (2003). Affordances as properties of the animal–environment system. *Ecological Psychology*, 15(2), 115–34.
- Striepens, N., Matusch, A., Kendrick, K. M., Mihov, Y., Elmenhorst, D., Becker, B., ... Bauer, A. (2014). Oxytocin enhances attractiveness of unfamiliar female faces independent of the dopamine reward system. *Psychoneuroendocrinology*, 39, 74–87.
- Stuchlik, A. (2014). Dynamic learning and memory, synaptic plasticity and neurogenesis: An update. *Frontiers in Behavioral Neuroscience*, 8(Article 106), 1–6. doi: 10.3389/fnbeh.2014.00106.
- Sweller, J., Ayres, P. L., & Kalyuga, S. (2011). *Cognitive load theory*. New York, NY: Springer. doi: 10.1007/978-1-4419-8126-4.
- Thorndike, E. L. (1913). *The psychology of learning*. New York, NY: Mason-Henry Press.
- Tolman, E. C. (1932). *Purposive behavior in animal and men*. New York, NY: The Century Company.
- Turk-Browne, N. B., Scholl, B. J., Chun, M. M., & Johnson, M. K. (2008). Neural evidence of statistical learning: Efficient detection of visual regularities without awareness. *Journal of Cognitive Neuroscience*, 21, 1934–45.
- Vuilleumier, P. (2005). How brains beware: Neural mechanisms of emotional attention. *Trends in Cognitive Science*, 9, 585–94.
- Vuilleumier, P. (2015). Affective and motivational control of vision. *Current Opinion in Neurology*, 28, 29–35.
- Vygotsky, L. S. (1980). *Mind in society: The development of higher psychological processes*. Cambridge, MA: Harvard University Press.
- Wang, Y., Fariello, G., Gavrilova, M. L., Kinsner, W., Mizoguchi, F., Patel, S., ... Tsumoto, S. (2013). Perspectives on cognitive computers and knowledge processors. *International Journal of Cognitive Informatics and Natural Intelligence*, 7(3), 1–24. doi: 10.4018/ijcini.2013070101.
- Warner, L. S. (2006). Native ways of knowing: Let me count the ways. *Canadian Journal of Native Education*, 29(2), 149.
- Weinstein, C. E., Husman, J., & Dierking, D. R. (2000). Interventions with a focus on learning strategies. In M. Boekaerts, P. R. Pintrich, & M. Zeidner (Eds.), *Handbook of self-regulation* (pp. 727–47). San Diego, CA: Academic Press.
- Weisz, J. R. & Stipek, D. J. (1982). Competence, contingency, and the development of perceived control. *Human Development*, 25, 250–81.
- Yin, H. H., Ostlund, S. B., & Balleine, B. W. (2008). Reward-guided learning beyond dopamine in the nucleus accumbens: The integrative functions of cortico-basal ganglia networks. *European Journal of Neuroscience*, 28(8), 1437–48. doi: 10.1111/j.1460-9568.2008.06422.x.